Layer stack, impedance and routing constraints

USBGE-ENG-002 | Document revision 04 | 2026-10-16

Design baseline USBGE-A-WT-20261016-R04 | A (native schematic title revision A0)

Layer construction

Order / layer	Material	Thickness mm	Dk
0 / Top Paste	Artwork		
1 / Top Overlay	Artwork		
2 / Top Solder	Solder Resist	0.01	3.500
3 / Top Layer	Copper	0.035001	
4 / Dielectric 2	IS400	0.12	4.200
5 / GND_1	Copper	0.035001	
6 / Dielectric 1	IS400	1.13	4.400
7 / GND_2	Copper	0.035001	
8 / Dielectric 3	IS400	0.12	4.200
9 / Bottom Layer	Copper	0.035001	
10 / Bottom Solder	Solder Resist	0.01	3.500
11 / Bottom Overlay	Artwork		
12 / Bottom Paste	Artwork		

Native copper thickness is 1.378 mil, approximately 0.035 mm on all four copper layers. Dielectrics are 0.120 / 1.130 / 0.120 mm; masks are approximately 0.010 mm each. Total nominal modeled thickness is 1.530 mm. The existing fabrication drawing rounds copper to 0.04 mm and separately states 1.55 mm finished thickness. No tolerance resolves the build definition; RR-02 controls agreement. GND_1 and GND_2 are native internal-plane layer names.

Controlled impedance

USB profile D90: differential 90 ohm +/-10%; Ethernet profile D100: differential 100 ohm +/-10%. These are native profile requirements, not measurements. Request supplier stack calculation and coupon/TDR reports at the agreed process revision. Changes to material, dielectric spacing, copper build or routing geometry require engineering disposition.

The profiles drive the differential rules. Native per-layer widths and the rule gap fields are recorded below. Profile solvers may override interactive geometry; no claim is made that every routed segment uses the preferred width.

Class	Impedance ohm	Top min / preferred / max width mm	Gap mm
DP_USB2_D90	90.000000	0.1000/0.1500/0.2462	0.1270
DP_USB3_D90	90.000000	0.1000/0.1500/0.2462	0.1270
DP_ETH_D100	100.000000	0.1200/0.1308/0.1839	0.1270

Matched lengths and routing layers

Rule	Tolerance mm	Scope
MATCH_ETH_BETWEEN_PAIRS	10.0000	InDifferentialPairClass('ETH_DIFF')
MATCH_ETH_WITHIN_PAIR	1.2700	InDifferentialPairClass('ETH_DIFF')
MATCH_USB3_WITHIN_PAIR	1.0000	InDifferentialPairClass('USB3_DIFF')

USB3 within-pair tolerance is 1.00 mm; Ethernet within-pair 1.27 mm; Ethernet between-pair 10.00 mm. A configured tolerance is not a measured residual skew. USB2 has no separately identified matched-length rule. Native net and pair classes are exported in USBGE-ENG-005.

USB2, USB3, Ethernet and clock routing have scoped routing-layer rules. Preserve their exact flags and priorities in USBGE-ENG-003; changing enabled layers is a design change. The full native rule parameters are also retained in Tools/design_extraction.json.

Fabrication-relevant rule limits

The active generic clearance is approximately 0.100 mm; USB-to-U2 breakout exception is approximately 0.080 mm. Effective board-edge object matrix is 0 mm; actual digital minimum0.250933 mm is assessed in ENG-014. The generic0.254 mm field is not enforcement. Minimum drilled hole is 0.300 mm. Disabled clearance rules are not requirements proven by the historical DRC.

USBGE-ENG-003 contains all 69 rule records, including enable state, priority, scope and native parameters. Minimum annular-ring, mask-sliver, paste, silk and component-clearance records must be included in the native DRC and fabricator DFM review. The historical drawing reports a 0.150 mm annular-ring limit and 0.075 mm mask-sliver limit; use the current rule export for baseline authority.